

CLOUD-07: CloudWatch Log Ingestion

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Overview

This notebook covers strategies for forwarding cloud provider logs into Dynatrace. While the focus is on AWS CloudWatch log forwarding (the most common pattern), the principles apply to Azure (Event Hub) and GCP (Pub/Sub) as well. You will learn about Amazon Data Firehose (the recommended approach), alternative methods, filtering strategies, OpenPipeline processing, and cost optimization.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	<code>logs.read</code> , <code>logs.ingest</code> , <code>openpipeline.read</code>
AWS Integration	CloudWatch log groups generating data
Prior Knowledge	CLOUD-01 and CLOUD-02 fundamentals

1. Log Ingestion Architecture

AWS CloudWatch Log Forwarding Options

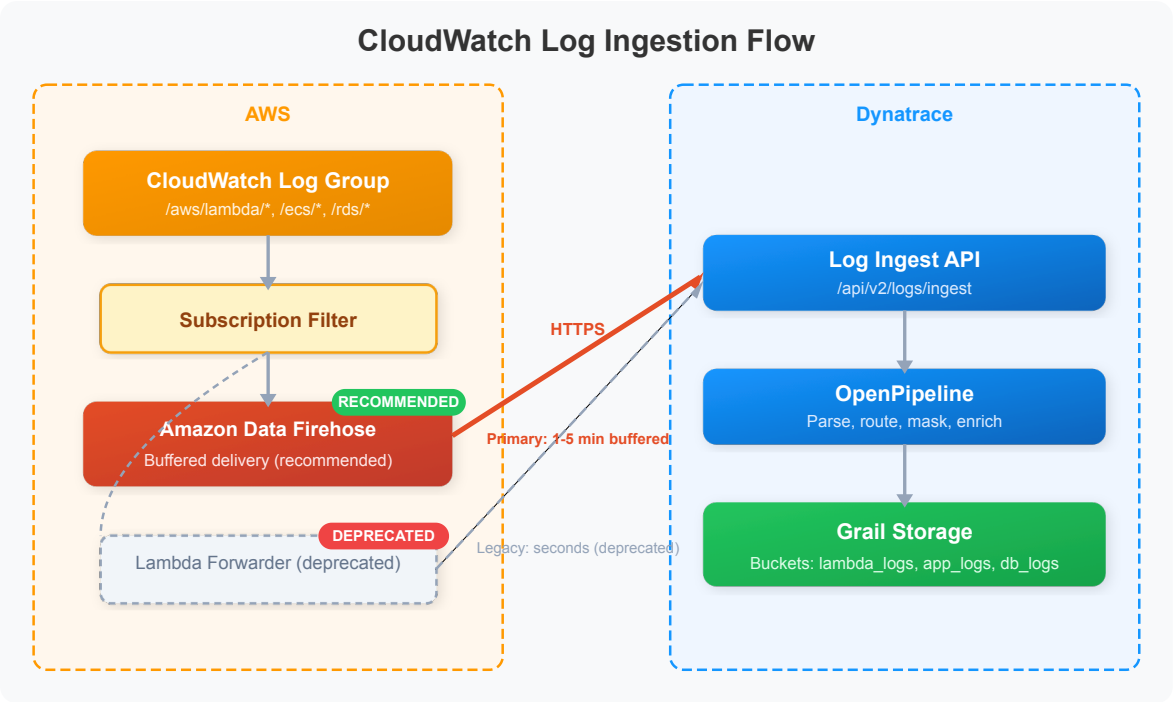
Method	Mechanism	Latency	Status
Amazon Data Firehose	CloudWatch → Firehose → Dynatrace API	1-5 min (buffered)	Recommended
S3 Ingestion	CloudWatch → S3 → Dynatrace serverless architecture	Minutes	Supported
Lambda Log Collection	Direct Lambda function log forwarding	Seconds	Supported (Lambda-specific)
Lambda Forwarder (Legacy)	CloudWatch Subscription Filter → Lambda → Dynatrace API	Seconds	Deprecated — migrate to Firehose
OpenTelemetry Collector	FluentBit/FluentD → OTLP → Dynatrace	Seconds	Supported (agent-based)

Important: The Dynatrace AWS Log Forwarder (Lambda-based) is **deprecated** and will not receive further updates. For new deployments, use **Amazon Data Firehose**. Existing Lambda forwarder deployments should plan migration to Firehose.

Cross-Cloud Log Forwarding

Cloud Provider	Primary Method	Alternative
AWS	Amazon Data Firehose	S3 ingestion, Lambda log collection
Azure	Azure Event Hub / Diagnostic Settings	Azure Functions forwarder
GCP	Pub/Sub via GKE integration	Cloud Functions + Pub/Sub

Architecture Overview



2. Amazon Data Firehose (Recommended)

Amazon Data Firehose is the recommended method for forwarding CloudWatch logs to Dynatrace. It provides a fully managed, serverless log pipeline with no infrastructure to maintain.

How It Works

1. **CloudWatch Subscription Filters** are attached to target log groups
2. Logs matching the filter are delivered to an **Amazon Data Firehose** delivery stream
3. Firehose buffers, compresses, and delivers logs to the **Dynatrace Log Ingest API**
4. Logs are processed through **OpenPipeline** and stored in **Grail**

Key Benefits

Feature	Description
Fully managed	No Lambda functions or custom code to maintain
Auto-scaling	Handles any log volume automatically
Smart retry	Built-in retry with exponential backoff on delivery failures
Buffering	Configurable buffer size (1-128 MB) and interval (60-900 seconds)
Compression	Automatic GZIP compression reduces transfer costs
S3 backup	Optional backup of all or failed records to S3

Setup via Clouds App

When you onboard an AWS account via the **Clouds app**, CloudWatch log forwarding via Firehose can be configured as part of the guided setup. The CloudFormation template creates:

- Firehose delivery stream configured with Dynatrace endpoint
- IAM roles for Firehose to deliver to Dynatrace
- Secrets Manager entry for the Dynatrace API token
- Optional: CloudWatch subscription filters for selected log groups

Manual Setup Steps

1. Create a **Firehose delivery stream** with HTTP endpoint destination
2. Set the endpoint URL to your Dynatrace log ingest API: `https://{env-id}.live.dynatrace.com/api/v2/logs/ingest`
3. Configure an **access key** using a Dynatrace API token with `logs.ingest` scope

4. Set buffer conditions (recommended: 1 MB or 60 seconds)
5. Enable GZIP compression
6. Create **CloudWatch Subscription Filters** on target log groups pointing to the Firehose stream

Subscription Filter Patterns

CloudWatch subscription filters support pattern matching to pre-filter logs before forwarding:

Pattern	What It Matches
<code>""</code> (empty)	All log events
<code>"ERROR"</code>	Lines containing ERROR
<code>?"ERROR" ?"WARN"</code>	Lines containing ERROR or WARN
<code>{ \$.level = "error" }</code>	JSON logs where level is error
<code>[ip, id, user, timestamp, request, status_code >= 400]</code>	Space-delimited logs with status >= 400

3. Alternative Ingestion Methods

S3 Log Ingestion

For logs already stored in S3 (e.g., ALB access logs, CloudTrail, VPC Flow Logs):

Feature	Description
Mechanism	S3 event notification → serverless architecture → Dynatrace API
Best for	Historical log import, services that write directly to S3
Parsing	Out-of-the-box parsing for common AWS log formats
Multi-region	Supports cross-region and multi-account setups

Lambda Log Collection

For AWS Lambda function logs specifically, Dynatrace offers direct log collection:

Feature	Description
Mechanism	Dynatrace Lambda Layer captures logs directly
Best for	Lambda-specific monitoring (cost-optimized, lower latency)
Prerequisite	Dynatrace Lambda Layer already deployed for tracing
Advantage	No subscription filter or Firehose needed for Lambda logs

Legacy: Lambda Forwarder (Deprecated)

Deprecated: The `dynatrace-aws-log-forwarder` Lambda function is deprecated and will not receive further updates. Migrate existing deployments to Amazon Data Firehose.

The legacy Lambda forwarder used CloudWatch Subscription Filters to invoke a custom Lambda function that forwarded logs to Dynatrace. While it provided low-latency delivery, it required maintaining custom Lambda infrastructure.

Method Comparison

Method	Latency	Maintenance	Cost	Best For
Firehose	1-5 min	None (managed)	Firehose + data transfer	General CloudWatch logs (recommended)
S3 ingestion	Minutes	Low	S3 storage + transfer	Historical logs, S3-native services
Lambda log collection	Seconds	None	Included with Lambda Layer	Lambda function logs
Lambda forwarder	Seconds	High (deprecated)	Lambda invocations	Do not use for new deployments

4. Filtering at Source vs Dynatrace

A critical design decision is **where** to filter logs: at the cloud provider level or within Dynatrace.

Comparison

Aspect	Filter at Source (CloudWatch)	Filter in Dynatrace (OpenPipeline)
Cost	Reduces ingestion volume and DDU cost	Full volume ingested, higher DDU cost
Flexibility	Limited pattern matching	Full DQL-based processing
Data loss risk	Filtered logs are not recoverable	All logs retained (can filter at query time)
Latency	Immediate (before forwarding)	At ingestion (OpenPipeline processing)
Maintenance	Subscription filters per log group	Centralized OpenPipeline rules

Recommended Hybrid Approach

- Source-level (CloudWatch):** Filter out known noise (health checks, debug logs in production)
- OpenPipeline:** Enrich, parse, route, and selectively discard or downsample
- Query-time:** Use DQL filters for ad-hoc analysis

Common Source-Level Filters

Log Group	Filter Pattern	Rationale
/aws/lambda/*	?"ERROR" ?"WARN" ?"CRITICAL"	Skip DEBUG/INFO for Lambda
/ecs/service/*	"" (all)	Forward all application logs
/aws/rds/*	?"error" ?"slow query"	Focus on errors and slow queries
ALB access logs	[w1, w2, w3, w4, w5, w6, w7, status_code >= 400, ...]	Only 4xx/5xx responses

5. OpenPipeline Processing

Once cloud logs reach Dynatrace, OpenPipeline processes them before storage in Grail.

Common OpenPipeline Operations for Cloud Logs

Operation	Purpose	Example
Parse	Extract structured fields from unstructured logs	Parse JSON, extract IP addresses
Route	Send logs to specific Grail buckets	Route Lambda logs to <code>lambda_logs</code> bucket
Enrich	Add metadata fields	Add <code>cloud.provider: aws</code> , <code>environment: prod</code>
Filter/Drop	Discard noisy log lines	Drop health check entries
Transform	Modify field values	Normalize severity levels

Example Pipeline Logic

```
Incoming cloud logs
|
|─ Parse JSON content (if JSON)
|─ Extract log_group and log_stream from attributes
|─ Add cloud.provider = "aws"
|─ Route /aws/lambda/* → bucket: lambda_logs
|─ Route /ecs/* → bucket: application_logs
|─ Route /aws/rds/* → bucket: database_logs
|─ Drop if content matches health check pattern
```

6. Querying Cloud Logs

All CloudWatch-Sourced Logs

```
// Recent logs from AWS (last hour)
fetch logs, from:-1h
| filter isNotNull(log.source) and contains(log.source, "aws")
| fieldsKeep timestamp, content, log.source, loglevel
| sort timestamp desc
| limit 20
```

Log Volume by Source

```
// Log volume by source over the last 24 hours
fetch logs, from:-24h
| summarize log_count = count(), by:{log.source}
| sort log_count desc
| limit 15
```

Error Log Trend Over Time

```
// Error logs per hour over the last 24 hours
fetch logs, from:-24h
| filter loglevel == "ERROR"
| makeTimeseries error_count = count(), interval:1h
```

Top Error Messages

```
// Most frequent error log messages in the last 6 hours
fetch logs, from:-6h
| filter loglevel == "ERROR"
| summarize error_count = count(), by:{content}
| sort error_count desc
| limit 10
```

Log Volume by Kubernetes Namespace

```
// Log volume by Kubernetes namespace over the last 6 hours
fetch logs, from:-6h
| filter isNotNull(k8s.namespace.name)
| summarize log_count = count(), by:{k8s.namespace.name, loglevel}
| sort log_count desc
| limit 20
```

7. Cost Optimization

Log ingestion is typically the largest cost driver in cloud monitoring. Optimization is critical.

Cost Impact by Layer

Layer	Cost Type	Optimization Lever
CloudWatch	\$0.50/GB ingested + \$0.03/GB stored	Reduce log retention, filter at source
Firehose	Firehose data processing (\$0.029/GB)	Buffer settings, compression
Dynatrace	DDU per GB ingested	Source filtering, OpenPipeline drop rules
Grail storage	DDU per GB stored x retention	Bucket-level retention policies

Optimization Strategies (Ranked by Impact)

Strategy	Estimated Savings	Complexity
1. Drop debug/trace logs at source	40-60%	Low
2. Filter health checks at source	10-30%	Low
3. Route to low-retention buckets	20-40% storage	Medium
4. Downsample repetitive logs in OpenPipeline	10-20%	Medium
5. Use sampling for high-volume services	Variable	Medium
6. Compress log payloads	5-10% transfer	Low

Monitoring Your Log Costs

```
// Log volume trend by hour over the last 7 days
fetch logs, from:-7d
| makeTimeseries log_count = count(), interval:1h
```

```
// Top 10 log sources by volume over the last 24 hours
fetch logs, from:-24h
| summarize log_count = count(), by:{log.source}
| sort log_count desc
| limit 10
```

8. Summary and Next Steps

Key Takeaways

- **Amazon Data Firehose** is the recommended approach for AWS CloudWatch log ingestion — fully managed, auto-scaling, no custom code
- The legacy Lambda-based log forwarder is **deprecated** — migrate existing deployments to Firehose
- **Filter at the source** (CloudWatch subscription filters) to reduce costs before logs reach Dynatrace

- Use **OpenPipeline** for enrichment, routing, and fine-grained filtering within Dynatrace
- **Monitor log volume** regularly to prevent cost surprises
- Apply similar patterns for **Azure (Event Hub)** and **GCP (Pub/Sub)** log forwarding

Next Steps

- **CLOUD-08: Multi-Cloud Patterns** — Unified log analysis across providers
- See the **OPLOGS notebook series** for deep dive into OpenPipeline log processing
- See the **OPMIG notebook series** for migrating from classic logs to OpenPipeline

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